Statistics & Data Analysis

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Review: The real number system**

The function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b, where $a > 0$ and $b > 0$.

Given the function $f(x) = a \cdot b^x$, and the points $(1, 12)$ and $(3, 108)$. From the point $(1, 12)$, we have: $12 = a \cdot b^1 \quad (1)$ From the point $(3, 108)$, we have: $108 = a \cdot b^3 \quad (2)$ Divide equation (2) by equation (1): $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Since $b > 0$, we take the positive square root: $b = \sqrt{9} = 3$ Now substitute the value of b into equation (1): $12 = a \cdot 3^1$ $12 = 3a$ $a = \frac{12}{3} = 4$ Therefore, $a = 4$ and $b = 3$.

2 **[Foundational]** • **Target: Review: Integer operations**

The first four terms of an arithmetic sequence are $7, 11, 15, 19, \ldots$. Which expression represents the nth term, a_n, of the sequence for all integers $n \geq 1$?

An arithmetic sequence has the general explicit formula $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference. From the given terms: 1. The first term is $a_1 = 7$. 2. The common difference is $d = 11 - 7 = 4$. Substituting these values into the formula: $a_n = 7 + (n - 1)(4) = 7 + 4n - 4 = 4n + 3$ Testing the first two terms: - For $n = 1$: $a_1 = 4(1) + 3 = 7$ - For $n = 2$: $a_2 = 4(2) + 3 = 11$ Therefore, the correct choice is $a_n = 4n + 3$. Common Misconceptions:

3 **[Foundational]** • **Target: Review: Operations with rational numbers**

An arithmetic sequence is defined by its first term $a_1 = 7$ and its common difference d. The sum of the first n terms of this sequence is given by $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$. If the sum of the first 10 terms is $S_{10} = 115$, find the value of the common difference d.

We are given the first term $a_1 = 7$ and the sum of the first 10 terms $S_{10} = 115$. The formula for the sum of the first n terms of an arithmetic sequence is $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$. Substitute the given values into the formula: $115 = \frac{10}{2}(2(7) + (10 - 1)d)$ $115 = 5(14 + 9d)$ Divide both sides by 5: $\frac{115}{5} = 14 + 9d$ $23 = 14 + 9d$ Subtract 14 from both sides: $23 - 14 = 9d$ $9 = 9d$ Divide by 9: $d = \frac{9}{9}$ $d = 1$ Therefore, the common difference d is 1.

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4 [Foundational] • Target: Review: Exponents and radicals

The first three terms of an arithmetic sequence are given by $a_1 = 5$, $a_2 = 8$, and $a_3 = 11$. What is the 10th term of this sequence?

The given sequence is an arithmetic sequence with the first term $a_1 = 5$. The common difference d can be found by subtracting any term from its succeeding term: $d = a_2 - a_1 = 8 - 5 = 3$. The formula for the n-th term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. To find the 10th term ($n = 10$), we substitute the values: $a_{10} = 5 + (10 - 1) \times 3 = 5 + 9 \times 3 = 5 + 27 = 32$. Therefore, the 10th term is 32.

5 [Foundational] • Target: Review: Laws of exponents

Consider the arithmetic sequence where the first term is 5 and the common difference is 3. What is the 4th term of this sequence?

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference. In this problem, we are given: $a_1 = 5$ $d = 3$ We need to find the 4th term, so $n = 4$. Substituting these values into the formula: $a_4 = 5 + (4 - 1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$ Therefore, the 4th term of the sequence is 14.

6 [Foundational] • Target: Units resulting from a formula or graph

Simplify the expression: $\frac{2^5 \cdot 2^3}{2^4}$.

To simplify the expression $\frac{2^5 \cdot 2^3}{2^4}$, we use the properties of exponents. First, we combine the terms in the numerator by adding their exponents: $2^5 \cdot 2^3 = 2^{5+3} = 2^8$. Then, we divide this result by 2^4 by subtracting the exponents: $\frac{2^8}{2^4} = 2^{8-4} = 2^4$. Common errors might include adding exponents in the denominator or incorrectly applying the division rule.

7 [Foundational] • Target: Review: The real number system

Simplify the expression: $(3x^2y^3)^2$

To simplify $(3x^2y^3)^2$, we apply the power of a product rule $(ab)^m = a^m b^m$ and the power of a power rule $(a^m)^n = a^{mn}$. $(3x^2y^3)^2 = 3^2 \cdot (x^2)^2 \cdot (y^3)^2$ First, calculate 3^2: $3^2 = 9$ Next, apply the power of a power rule to the variables: $(x^2)^2 = x^{2 \times 2} = x^4$ $(y^3)^2 = y^{3 \times 2} = y^6$ Combining these results, we get: $9x^4y^6$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Review: Integer operations

A line passes through the points $(2, 3)$ and $(6, 11)$. What is the slope of this line?

The slope of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the points $(2, 3)$ and $(6, 11)$, let $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (6, 11)$. Substituting these values into the slope formula: $m = \frac{11 - 3}{6 - 2}$ $m = \frac{8}{4}$ $m = 2$ Therefore, the slope of the line is 2.

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9 [Intermediate] • Target: Review: Operations with rational numbers

What is the solution to the equation $2^{x+1} = 32$?

To solve the exponential equation $2^{x+1} = 32$, express both sides with the same base: 1. Write 32 as a power of 2: $32 = 2^5$ 2. Substitute this into the equation: $2^{x+1} = 2^5$ 3. Since the bases are equal and positive (and not equal to 1), equate the exponents: $x + 1 = 5$ 4. Subtract 1 from both sides: $x = 4$

10 [Intermediate] • Target: Review: Exponents and radicals

Simplify the expression: $(a^3)^2 \cdot a^4$

Using the power of a power rule $(x^m)^n = x^{m \cdot n}$ and the product of powers rule $x^m \cdot x^n = x^{m+n}$, we can simplify the expression. First, $(a^3)^2 = a^{3 \cdot 2} = a^6$. Then, multiply this by a^4: $a^6 \cdot a^4 = a^{6+4} = a^{10}$.

11 [Intermediate] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(5)$.

Final Answer: 25. Substitute $x = 5$: $f(5) = 4(5) + 5 = 25$.

12 [Intermediate] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

13 [Intermediate] • Target: Review: Exponents and radicals

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(2)$.

Final Answer: 16. Substitute $x = 2$: $f(2) = 5(2) + 6 = 16$.

14 [Intermediate] • Target: Review: Exponents and radicals

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 6.01 Systems & Inequalities

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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16 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(3)$.

Final Answer: 21. Substitute $x = 3$: $f(3) = 5(3) + 6 = 21$.

17 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

18 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 9.01 Radicals & Exponents

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(4)$.

Final Answer: 26. Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

19 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 1.01 Linear Equations & Lines

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

20 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(5)$.

Final Answer: 31. Substitute $x = 5$: $f(5) = 5(5) + 6 = 31$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

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22 [Foundational] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(5)$.

a. 13 ✓

b. 15

c. 10

d. 17

Substitute $x = 5$: $f(5) = 2(5) + 3 = 13$.

24 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

25 [Foundational] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

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27 [Foundational] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

28 [Foundational] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Review: The real number system

A data set recorded for review: the real number system consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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31 [Intermediate] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(3)$.

a. 9 ✓

b. 11

c. 6

d. 13

Substitute $x = 3$: $f(3) = 2(3) + 3 = 9$.

32 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

33 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

34 [Intermediate] • Target: Review: Integer operations

A data set recorded for review: integer operations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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35 [Intermediate] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

36 [Intermediate] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

37 [Intermediate] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for review: operations with rational numbers consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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39 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 1.01 Linear Equations & Lines

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(3)$.

a. 17 ✓

b. 21

c. 12

d. 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

40 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 10.01 Rational Expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

41 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

42 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 3.01 Exponents & Polynomials

A data set recorded for review: laws of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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43 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 6x + 7$ in review: laws of exponents, evaluate $f(3)$.

a. 25 ✓

b. 31

c. 18

d. 29

Substitute $x = 3$: $f(3) = 6(3) + 7 = 25$.

44 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A data set recorded for units resulting from a formula or graph consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

45 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 6.01 Systems & Inequalities

For the model $f(x) = 2x + 8$ in units resulting from a formula or graph, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

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